GRACE-Mem: Graph Retrieval with Agentic Curation of Evidence for Long-Term Conversational Memory

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Long-term answers need source evidence, not summaries

- **Information Loss**: Compression may omit speakers, temporal qualifiers, or critical details.
- **Insufficient Evidence**: A relevant summary may not contain all the clues needed to answer.
- **GRACE-Mem's Approach: Retrieve structured memories, verify provenance-linked raw dialogue, and generate evidence-grounded answers.**

EXAMPLE

- **User Question**: “How old will I be when Rachel gets married?” (Sep 1, 2022)
- **Clue 1 — Wedding Timing**: “Rachel is getting married next year.” → 2023
- **Clue 2 — Current Age**: “I'm 32, so I'm in my 30s.” → 32 in 2022
- **Correct Answer**: 33 years old.

Motivation

- **Long-term conversations:** The dialogue history can exceed the model context window, so assistants need external memory and retrieval.
- **The evidence problem:** Compact summaries, facts, and graph records are useful indexes, but may omit speakers, temporal qualifiers, or conditions.
- **Core goal:** Retrieve the right source-level dialogue evidence and keep the final answer grounded in verifiable conversation turns.
- **GRACE-Mem:** Use structured memory to precisely locate clues, but use provenance-linked raw dialogue as the basis for the final answer.

Related Work

- **Summarization & compression:** MemGPT, SeCom, LangMem, and related systems store compact representations and retrieve them on demand, but the compression process can lose important details such as speakers and temporal conditions, and may even introduce unsupported inferences.
- **Structured memory:** A-MEM, Zep, and Mem0 connect entities, relations, and memory units to support evolving long-term memory, but semantic similarity does not necessarily indicate that a memory contains the evidence needed to answer a question.
- **Graph-aware retrieval:**LightRAG, MRAgent, and SA-RAG leverage graph structures to connect evidence beyond flat similarity search and support multi-hop retrieval, yet retrieved memory representations are primarily useful for locating relevant clues rather than serving as authoritative evidence for the final answer.

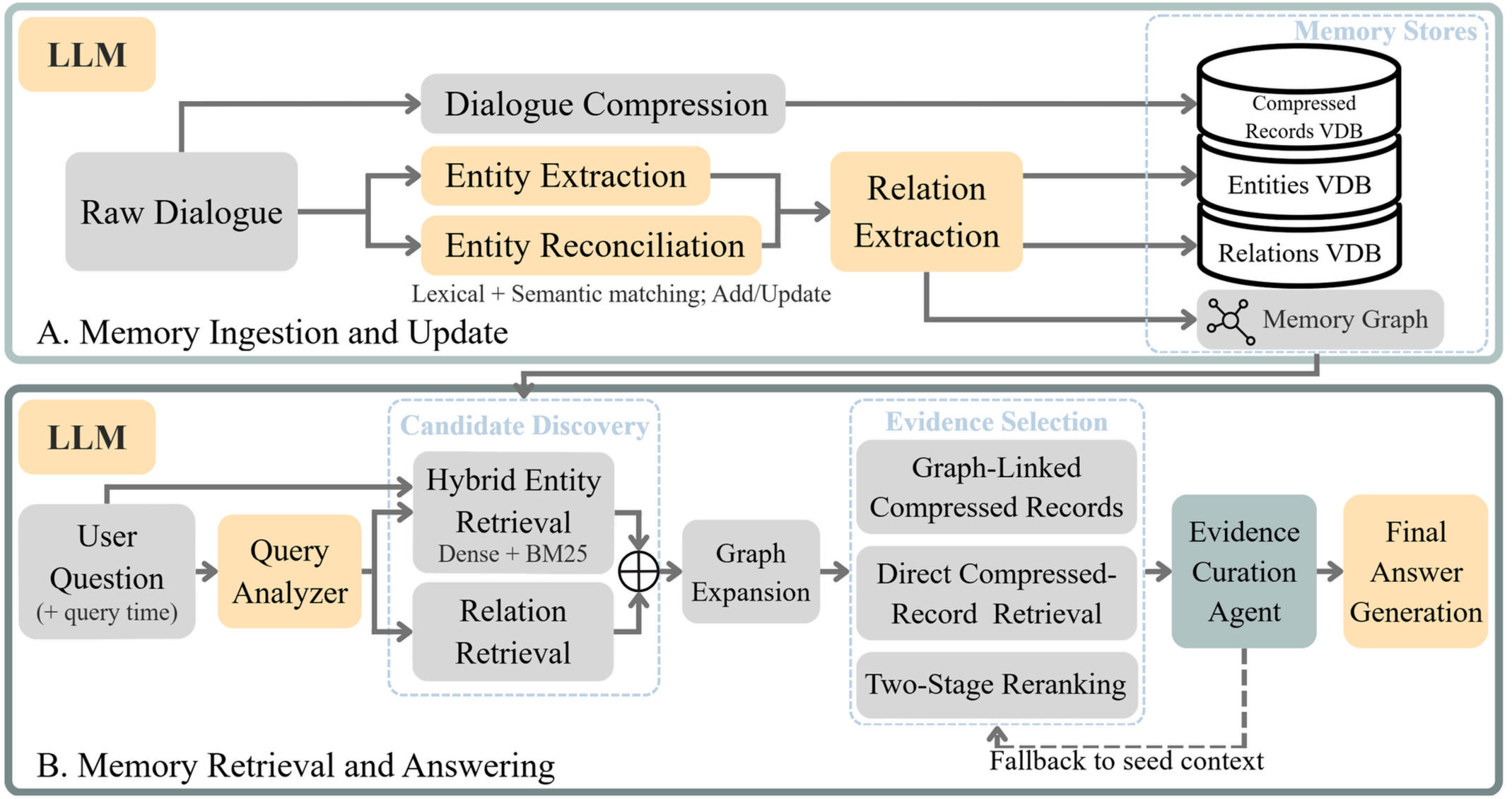
Contributions

1. **Source-evidence-centric role-decoupled architecture:** Structured memory serves only as a navigational index; final answers are grounded exclusively in provenance-linked raw dialogue.

2. **Complementary retrieval and hierarchical reranking:** Combine graph expansion with direct compressed-vector retrieval, then apply two-stage reranking at the graph and record levels to improve answer-relevant coverage and precision.

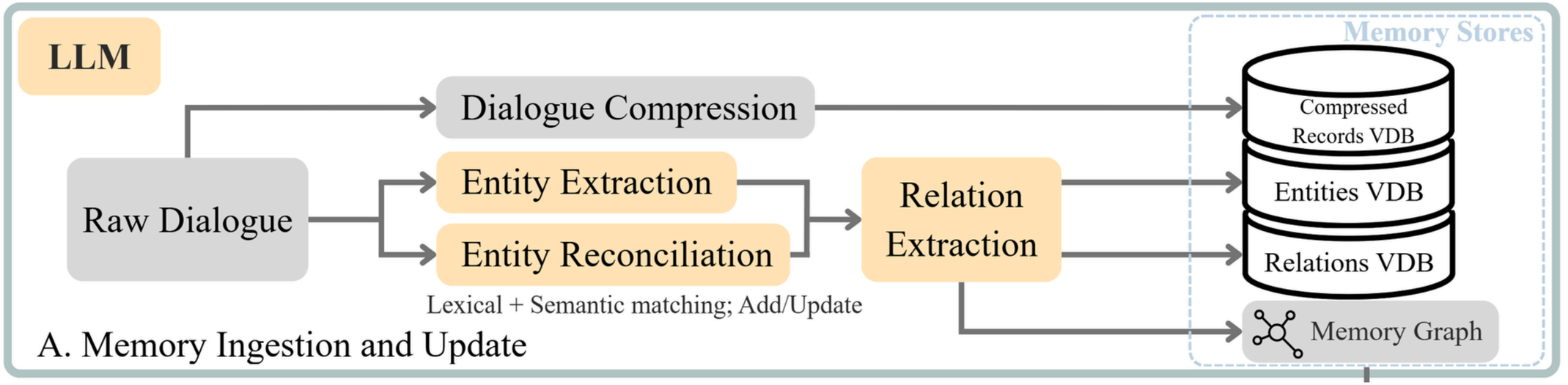
3. **Post-retrieval agentic Evidence Curation:** A lightweight Agent invokes GREP / READ / VEC to actively verify evidence, filter distractors, and recover missing clues, with identity fallback for robustness.

Pipeline


LLM
Dialogue Compression
Raw Dialogue
Entity Extraction
Entity Reconciliation
Lexical + Semantic matching; Add/Update
Relation Extraction
Memory Stores
Compressed Records VDB
Entities VDB
Relations VDB
Memory Graph
A. Memory Ingestion and Update
LLM
Candidate Discovery
User Question (+ query time)
Query Analyzer
Hybrid Entity Retrieval Dense + BM25
Relation Retrieval
Graph Expansion
Evidence Selection
Graph-Linked Compressed Records
Direct Compressed-Record Retrieval
Two-Stage Reranking
Evidence Curation Agent
Final Answer Generation
Fallback to seed context
B. Memory Retrieval and Answering

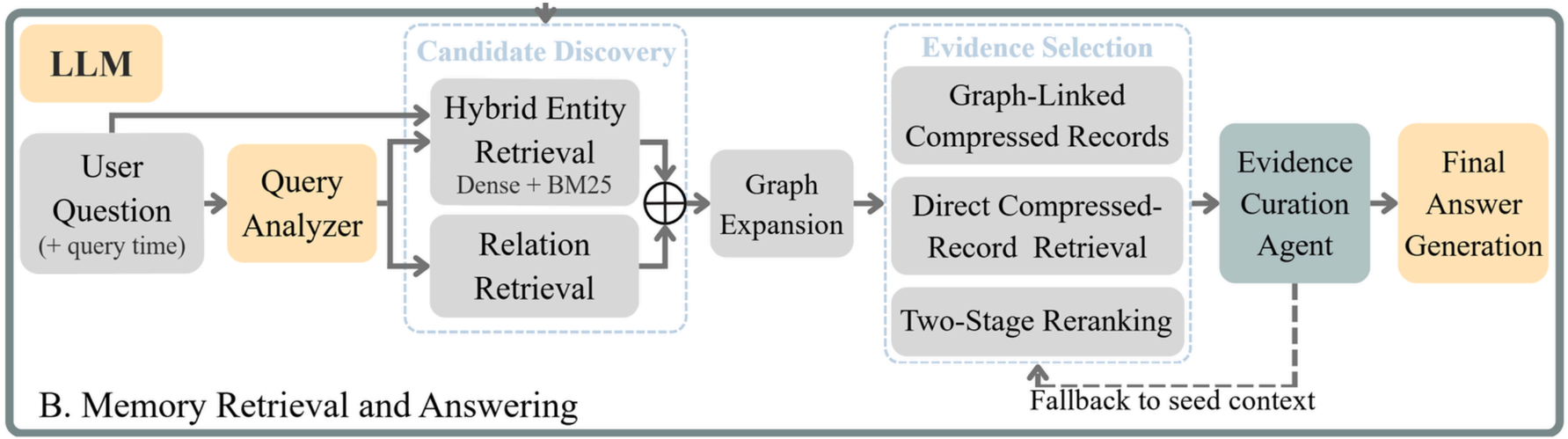
Design Step I — Memory Ingestion

- **Compress dialogue:** Each dialogue unit becomes a compact record for retrieval.
- **Extract entities:** The system identifies entities and resolves them against existing memory using lexical + semantic matching.
- **Extract relations:** Relations are constructed only between reconciled entities.
- **Preserve provenance:** Every record keeps session/message identifiers that point back to the original dialogue span.



Design Step II — Retrieval, Reranking & Curation

- **Candidate discovery: Hybrid Entity + Relation Retrieval → Graph Expansion**
  From the query, identify relevant entities and relations, then explore potentially relevant memories through graph expansion.
- **Evidence selection: Graph-linked + Direct Retrieval → Two-Stage Reranking**
  Combine graph-linked records with direct retrieval results and select Top-K candidates through two-stage reranking.
- **Raw evidence: Compressed Records → Raw Dialogue Evidence**
  The Agent returns to the raw dialogue based on candidate records to verify, supplement, and filter evidence that truly supports the answer.



Evidence Curation Agent: From Candidates to Verifiable Evidence

1 Query + Seed Evidence

User question
(2022/09/01 23:52)
How many years will I be when my friend Rachel gets married?

Seed evidence
16 seed turns
- Contains clue about Rachel's wedding ("next year").
- No user age information.

Key seed
answer_cbd08e3c_1:2:u
(2022/09/01 03:06)
User: "my friend Rachel's getting married next year... "

2 Agent Reasoning + Tool Actions

Evidence Curation Agent

0 **GREP** (age|born|birthday|years old)
Result: 170 matches (too broad)

1 **GREP** (I am \d+|born in \d{4}|age \d+)
Result: 0 matches

2 **VECTOR** (user age birthday born year)
Weak or irrelevant results.

3 **READ** answer_cbd08e3c_1:2:u
Confirm: Rachel's wedding is "next year"

• • •

k **GREP** (I am \d+|born in \d{4}|age \d+)
Result: 1 matches → **answer_cbd08e3c_2:4:u**

k+1 **READ** answer_cbd08e3c_2:4:u
Confirm: "I'm 32, so I'm in my 30s."

k+2 **FINAL**
answer_cbd08e3c_1:2:u,
answer_cbd08e3c_2:4:u

3 Evidence Validation

Agent re-checks the 15 seed turns that were dropped in Stage 2 and keeps any that provide missing information.

0 turns are age facts

1 turn is age-related but not a fact
(asks user's age range, not actual age)

13 turns are unrelated
(apps, to-do list, streaming, alumni, language exchange, etc.)

1 turn about Rachel's wedding
(already have a better seed)

No additional seed is useful.
Proceed with the two selected turns from Stage 2.

4 Final Curated Context & Answer

Final curated context (2 turns)

answer_cbd08e3c_1:2:u
(2022/09/01 03:06)
User: " ... my friend Rachel's getting married next year ... "

answer_cbd08e3c_2:4:u
(2022/09/01 16:14)
User: "I'm 32, so I'm in my 30s."
Agent Add

Reasoning
1. "Next year" → 2023 (relative to 2022/09/01)
2. User is 32 in 2022 → will be 33 in 2023

Answer:
You will be 33 years old.

Benchmarks

- **LoCoMo (1,540 questions):**
  - Long, multi-session conversations between two fictional personas, with evolving events and relationships.(~600 turns, ~16K tokens).
  - Tasks: Single-hop, Multi-hop, Temporal, and Open-domain reasoning.

- **LongMemEval-S (500 questions):**
  - Long, timestamped user-assistant interaction histories containing relevant sessions and distractors.(~40 sessions, ~115K tokens).
  - Tasks: Single-session recall, Multi-session reasoning, Temporal reasoning, and Knowledge update.

Main Results

- GRACE-Mem achieves overall performance above existing baselines on both LoCoMo and LongMemEval.

- The gains do not rely on a single answer backbone; strong performance is maintained across all three backbones, indicating that the overall improvement comes from the retrieval and evidence-processing architecture.

Model / Baseline	LoCoMo Overall	LongMemEval Overall
Published memory systems (Backbone: GPT-4o-mini)		
RAG (Lewis et al., 2020)	30.2	37.4
MemGPT (Packer et al., 2023)	41.5	55.4
Zep (Rasmussen et al., 2025)	58.5	63.2
A-MEM (Xu et al., 2025)	52.5	53.8
Mem0 (Chhikara et al., 2025)	65.3	57.4
Nemori (Ma et al., 2026)	73.0	64.2
AMA (Huang et al., 2026)	77.4	69.8
LangMem (LangChain, 2026)	51.5	52.8
GRACE-Mem		
OSS-20B	84.6	77.6
OSS-120B	**86.9**	**80.2**
GPT-4o-mini	85.3	73.2

Same GPT-4o-mini backbone as baselines: **+7.9 / +3.4 pp over AMA**

Effect of the Agent Curation of Evidence

- On LongMemEval, adding filter-fetch improves all three models, and adding evidence validation yields further gains, indicating that validation helps confirm the evidence found by the Agent.
- LongMemEval shows the largest improvement, with OSS-120B increasing from **76.0% → 79.0% → 80.2%**, the largest gain among the three models.
- On LoCoMo the effect is mixed: for OSS-120B the no-agent baseline (87.3%) is already the best. When retrieval is sufficient, filtering can remove useful context, which motivates adaptive curation.

	LongMemEval			LoCoMo		
Configuration	**OSS-20B**	**OSS-120B**	**GPT-4o-mini**	**OSS-20B**	**OSS-120B**	**GPT-4o-mini**
Baseline	76.2	76.0	69.2	84.0	**87.3**	84.2
filter-fetch	77.2	79.0	**71.8**	82.3	86.1	83.6
filter-fetch + evidence validation	**77.6**	**80.2**	73.2	**84.6**	86.9	**85.3**

Ablation Study

- No Graph: Graph structure is important for linked evidence discovery.
- No Direct Retrieval: Recovers evidence missed by the graph.
- Compressed Only: Compressed representations cannot fully replace source dialogue.
- Conclusion: Graph Discovery, Direct Coverage, and Raw Evidence are highly complementary roles.

Ablation	**LoCoMo**	**LongMemEval**
Full Pipeline	84.0	76.2
No Graph	66.4 (**−17.6)**	67.9 (−8.3)
No Direct Retrieval	80.2 (−3.8)	69.7 (**−6.5**)
Compressed Only	76.2 (**−7.8**)	70.0 (−6.2)

Conclusions & Future Directions

Core Conclusions:

1. Structured memories serve as retrieval signposts, while raw dialogue remains the authoritative source of evidence.
2. GRACE-Mem combines complementary retrieval and evidence verification mechanisms to improve long-term conversational question answering.
3. Memory systems should not be judged by how much memory they retrieve, but by whether the right evidence survives to the answering stage.

Limitations & Future Directions:

1. Adaptive Agent Curation: Dynamically decide whether to call Agent, avoid over-filtering when retrieval is already saturated.
2. Explicit temporal graph evolution: Strengthen version control for changing facts.

Thank You — Questions?

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Scan the QR code →

CODE

github.com/gbtailab-tech/GRACE-Mem